

## Part I

# Critical note reading

This section includes notes from class lectures and discussions, with intended or unintended alterations, incorrect statements, or typos. Critical reading is the active process of learning, analysing, interpreting, evaluating, and synthesizing the claims in the text rather than accepting them as stated. More importantly, we can modify, extend certain claims, or make new and better ones.

## 1 Graph representation of data

### 1.1 Datum-to-vertex mapping: $X \rightarrow V$

Consider a collection  $X$  of datum entries,

$$X = \{x_j \mid j = 1 : n_o\}$$

where  $x_j$  is the attribute descriptor for datum entity/entry indexed by  $j$ .

- ☐ Convert the (raw) attribute descriptors  $x_j$  (or a portion of which that is under consideration) to a feature vector  $f_j$  in a metric space of certain dimension  $d \geq 1$ ,  $j = 1 : n_o$ .
- ☐ The attribute-to-feature mapping partitions  $X$  into  $n$  feature-equivalent subsets,  $n \leq n_o$ .

The mapping  $X \rightarrow V$ : every vertex represents a unique feature-equivalent data subset.

1.1.1 The (raw) attribute descriptor  $x_j$  may contain more than one subfields (or subdomains), each is of a different data type. Give an example for each type:

- categorical:
- logical values/expressions:
- numerical values:
- statistical quantities/measures:

1.1.2 [Y/N/M]

- Every attribute subfield under consideration is to be translated to a numerical feature subvector in a metric (or pseudo-metric) space of a chosen dimension.
- For non-numerical attributes, an attribute-to-feature translation is necessary;
- For numerical attributes, an attribute-to-feature transform is frequently carried out to make the primary information salient and filter out other components, especially, wanted noise.
- The number of feature dimensions is necessarily specified, but often with the absence of supporting arguments (lack of transparency).

1.1.3 [Y/N/M]

The mapping from attributes to a feature vector is not necessarily reversible.

- The set of datum entries with the same feature vector is the inverse image of the feature vector; the datum entries indistinguishable by their feature description are considered in the same equivalence class/cluster.
- The dataset  $X$  is partitioned by the feature extraction. The mapping between the feature vectors and their inverse image sets are 1-to-1.
- The set of datum-vertices is 'contracted' to the set of feature inverse images.

#### 1.1.4 [Y/N/M]

For a digital or digitized image of  $n_1 \times n_2$  pixels in one color channel (R, G, B or gray), one may use any one of the following feature descriptors:

- intensity value descriptor over the pixel grid (domain);
- horizontal and vertical gradients over the pixel grid;
- HOG descriptor over a convoluted grid of block subimages.

Here, HOG stands for Histogram of Oriented Gradients (introduced in class).

#### 1.1.5 [Y/N/M]

The HOG feature is a stochastic descriptor, it has the following advantages:

- maintaining the advantages of gradients: capturing shape information, invariant to shifts in the image intensity;
- not sensitive to *intra-variation* in gradients due to random noise in immediate neighbor pixels;
- not sensitive to *inter-variation* across cell blocks.

*Optional.* Provide a brief description how to apply the spatial encoding approach used for constructing HOG features to a case in the following:

- 1-a Orientation-Magnitude encoding: the magnitude information is important to certain down-stream analysis tasks;
- 1-b Temporal encoding:  $x_j$  is a time sequence, a dynamic signal;
- 1-c Space-Time encoding:  $x_j$  is a video sequence;
- 1-d A specific large dataset you are particularly interested in.

## 1.2 Basic operations and combinatorial measures

#### 1.2-a [Y/N/M]

A graph is strongly connected

- if and only if there is no isolated nodes;
- if and only if every pair of nodes is connected by a path;
- if and only if the diameter is finite;
- if and only if  $A$  is irreducible.

A directed graph (digraph) is weakly connected if its undirected counterpart is strongly connected.

1.2-b Name two well known algorithms for finding the number of strongly and weakly connected components.

1.2-c [Y/N/M]

A network is highly vulnerable if it is disconnected by the removal of a single node or edge. See the following two different cases:

- A tree network is disconnected by the removal of any non-leaf node.
- A  $d$ -dimensional hypercube network remains well connected upon removal of any single node or edge.

1.2-d [Y/N/M]

Denote by  $A$  the adjacency matrix of a directed graph  $G = (V, E)$ , each edge  $(i, j) \in E$  is an ordered 2-tuple and weighted by  $w_{ij}$ ,

$$A = \sum_{(i,j) \in E} w_{ij} (e_i e_j^\top) \quad (1)$$

and

$$A = \sum_{i \in V} \sum_{j \in \mathcal{N}(i)} w_{ij} (e_i e_j^\top) = \sum_{i \in V} S_i \quad (2)$$

where  $S_i$  is the (directed) star graph centered at node  $i$  with edges to its neighbors:

$$S_i = (V_i, E_i), \quad V_i = \mathcal{N}[i], \quad E_i = E \cap (\{i\} \times V)$$

corresponding to the row- $i$  of  $A$ .

For an undirected graph  $G$ , each edge  $(i, j)$  is an unordered tuple,

$$A = \sum_{(i,j) \in E} w_{ij} (e_i e_j^\top + e_j e_i^\top) \quad (3)$$

1.2-e Let  $G$  be a digraph. For its adjacency matrix representation, we use the following convention on edge directions:

- column- $j$  of  $A$  contains the outgoing edges from node  $j$ ,
- row- $j$  of  $A$  contains the incoming edges to node  $j$ .

[Y/N/M]

The in-degree vector and the out-degree vector can be calculated via matrix-vector multiplications:

$$d_{\text{in}} = Ae, \quad d_{\text{out}} = A^\top e, \quad (4)$$

Each vector represents a scalar function on the vertex set  $V$ .

- A source node has zero in-coming degree.
- A sink node has zero outgoing degree.

- Let  $G$  be a citation graph among a set of published articles. With the citing and cited articles indexed according to publication time, the most recent indexed the last, the adjacency matrix is nearly upper triangular.

1.2-f [Y/N/M]

Let  $H$  be the subgraph induced by a vertex subset  $V_h$ ,  $V_h \subset V$ . Then, the adjacency matrix of  $H$  is the submatrix of  $A$  with the rows and columns associated with  $V_h$ :

$$A(H) = A(V_h, V_h)$$

The adjacency of any subgraph with vertex set  $V_h$  has its elements in  $A(H)$ .

1.2-g [Y/N/M]

Let  $G = (V, E)$  be undirected and unweighted, represented by its adjacency matrix  $A$ . Let  $G_2 = (V, E_2)$  be the graph such that  $(i, j) \in E_2$  if and only if node  $i$  and node  $j$ ,  $i \neq j$ , have common neighbors, and  $(i, j)$  is weighted by the number of their common neighbors. Then,

$$A_2 \triangleq A(G_2) = A^2 - \text{diag}(A^2), \quad \text{diag}(A^2) = Ae = d(G). \quad (5)$$

- $A(i, :)$  is the indicator of nodes adjacent to node  $i$ ;  $\text{nnz}(A(i, :))$  is the number of nodes reached from  $i$  by one step.
- $\text{nnz}(A_2(i, :))$  is the number of nodes reached from  $i$  by exactly two steps.
- $\text{nnz}(A + A_2)(i, :)$  is the number of nodes in 2-leg neighborhood of  $i$ .

1.2-h Let  $G = (V, E)$  be undirected and unweighted, with adjacency matrix  $A$ .

- o Give an interpretation of the edges in graph  $G_3 = (V, E_3)$  represented by  $A_3 = A^2 \odot A$ , where  $\odot$  denotes Hadamard product.
- o Relate the degrees on  $G_3$ ,  $d_3 = A_3 e$ , to the numbers of triangles on  $G$ , and then to the local cluster coefficients on  $G$ . Given a simple formula.

1.2-i *Optional*.

Let graph  $G$  be undirected and unweighted, with adjacency matrix  $A$ .

Give an interpretation for each of the following graphs, for  $k > 2$ ,

- o  $A_k = A^k - \text{diag}(A^k)$ ,
- o  $R_k = \left( \sum_{j=1:k} A^j \right) - \text{diag} \left( \sum_{j=1:k} A^j \right)$ .

### 1.3 Construction of Near-Neighbor(NN) Graph(s)

- Start with all-to-all pairwise relationships from vertex feature vectors:
  - geometric: similarity, Euclidean distance
  - stochastic: divergence, mutual information

In each case, the all-to-all pairwise relationships typically result in a full interaction matrix.

- Why NN graphs? Often, the encoded feature points (vectors) are in a high-dimensional Euclidean space  $\mathbb{R}^d$ ,  $n \ll 2^d$ . They are believed to reside on a latent non-Euclidean/Riemannian manifold, or simply put, in a low-dimensional curved space, or a parameterized space with a significantly small set of parameters.

A near-neighbor (NN) graph is used as a primitive and discrete descriptor of the latent manifold, by empirical sample points on the manifold. It has the two key properties:

- preserving local Euclidean geometry and local neighborhood;
- maintaining neighborhood-neighborhood connections globally.

and global geodesic distances, and it is used to reveal or uncover the latent curved feature space.

There are several ways to construct near-neighbor graphs from all-to-all pairwise relationships:

- Hard cut-off : e.g. RNN by neighbor radius<sup>1</sup>, KNN by neighbor population;
  - Soft fall-off : e.g. conditional neighbor distributions with a parameter for perplexity.
- The hyperparameter(s) (radius- $r$ , population- $k$  and perplexity) for these NN graphs are fundamentally related to the unknown dimension of the latent curved space and the data sample distribution on the space. They are to be determined empirically in various ways.

---

<sup>1</sup>If the feature vectors are in high-dimensional space, this approach suffers from a high-dimensional curse,

- 1.3.a. Give an example that the feature points are on a curved space embedded in an Euclidean space, for a specified radius  $r$ ,  $r > 0$ , the neighbors of a feature point with  $r$ -neighborhood by geodesic distance are not the same as those by the Euclidean distance.
- 1.3.b. Give an example how to measure pairwise similarity or discrepancy for each of the following cases
- The feature vectors are in a geometric metric space
  - The feature vectors are non-negative and stochastic
- 1.3.c. Describe briefly how a sorting procedure is used for constructing a near-neighbor graph,  $G_{\text{rnn}}$  or  $G_{\text{knn}}$ , by a hard cut-off threshold.
- 1.3.d. [Y/N/M]
- The adjacency matrix of a RNN  $G_{\text{rnn}}$  is symmetric,  $A^\top = A$ .
  - The adjacency matrix of a KNN  $G_{\text{knn}}$  is non-symmetric. Often, it is symmetrized by  $A := A \cup A^\top$ . Upon the symmetrization, an edge between  $i$  and  $j$  indicates that one of them is among the  $k$ -nearest-neighbors of the other.
- 1.3.e. [Y/N/M]
- For a RNN  $G_{\text{rnn}}$ ,  $A$  is uniformly sparse across local neighborhoods;
  - For a KNN  $G_{\text{knn}}$ ,  $A$  is uniformly sparse across local neighborhoods
- .

## 2 Graph Spectral Analysis

In this section, graph  $G$  is assumed undirected ( $A^\top = A$ ), nonnegatively weighted ( $w(E) > 0$ ), without self-loops ( $\text{diag}(A) = 0$ ), without isolated nodes ( $d > 0$ ).

### 2.1 The incidence matrix & the graph Laplacian

□ The vertex-edge incidence matrix is  $B = B(V, E)$ :

$$B(:, (i, j)) = \pm(e_i - e_j), \quad (i, j) \in E. \quad (6)$$

□ The ordinary Laplacian is

$$L = B \text{diag}(w(E)) B^\top, \quad (7)$$

where  $W(E)$  is diagonal with the edge weights as the diagonal elements. If  $G$  is unweighted,  $W(E)$  is the identity matrix  $I_m$ ,  $m = |E|$ .

2.1.a [Y/N/M] The Laplacian is the accumulation of all edge Laplacians

$$L(G) = \sum_{(i,j) \in E} L((i, j)) = \sum_{(i,j) \in E} (e_i - e_j)(e_i - e_j)^\top \quad (8)$$

2.1.b [Y/N/M] The Rayleigh quotient of the Laplacian is

$$r(L, x) = x^\top L x = \sum_{(i,j) \in E} w_{ij} (x_i - x_j)^2, \quad x^\top x = 1. \quad (9)$$

2.1.c [Y/N/M] If  $G$  has  $k$  disconnected components  $H_j$ ,  $j = 1, \dots, k$ , then

$$L(G) = \bigoplus_{j=1:k} L(H_j), \quad r(L, x) = \sum_{j=1:k} r(L_j, x_j), \quad x^\top = [x_1^\top, \dots, x_k^\top].$$

2.1.d [Y/N/M]

- The graph Laplacian matrix is symmetric, semi-positive definite; all eigenvalues of  $L$  are real-valued and non-negative. Consequently, its eigenvalues can be ordered in non-descending order:

$$\lambda_{\min} = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n = \lambda_{\max}.$$

- $\text{diag}(L) = D = \text{diag}(Ae)$  and  $L = D - A$ .
- $\lambda_1 = 0$  and  $Le = 0$ .
- $\lambda_1 \leq d_{\min}$ ,  $\lambda_{\text{avg}} = d_{\text{avg}}$ ,  $\lambda_n \geq d_{\max}$ .
- $\lambda_2 > 0$  if and only if  $G$  is connected. The multiplicity of the zero eigenvalue is equal to the number of connected components.

2.1.e [Y/N/M] Denote by  $G_0$  the combinatorial unweighted counterpart of version of  $G$ . By Kirchhoff's matrix-tree theorem (1947),

$$\#\text{spanning-trees}(G) = \frac{1}{n} \prod_{\lambda_k \neq 0} \lambda_k(L_0). \quad (10)$$

## 2.2 The Random-walk Transition Matrix & the Normalized Laplacian

☐ The matrix  $P_c = AD^{-1}$  is column stochastic; the matrix  $P_r = D^{-1}A$  is row stochastic. Each is known as the random-walk transition matrix.

☐ The normalized Laplacian is

$$\hat{L} = D^{-1/2}LD^{-1/2} = I - \hat{A}, \quad \hat{A} = D^{-1/2}AD^{-1/2}. \quad (11)$$

2.2.a [Y/N/M] If  $G$  is regular with constant degree  $d$ , then  $\hat{L} = d \cdot L$ .

2.2.b Matrix  $\hat{A}$  is spectrally similar to  $P_c$  and to  $P_r$ .

2.2.c [Y/N/M]

The eigenvalues of  $\hat{L}$  are in  $[0, 2]$ , their average value is equal to 1.

2.2.d The (discrete) *Cheeger constant*  $h(G)$  is a combinatorial measure of the graph connectivity. Describe briefly its definition.

2.2.e [Y/N/M]

The Fiedler value  $\lambda_2$  is an algebraic measure of graph connectivity. It can be obtained efficiently quadratic minimization of the Laplace's Rayleigh-quotient. It is related to the Cheeger constant as follows

$$\frac{1}{2}h^2 \leq \lambda_2(\hat{L}) \leq 2h. \quad (12)$$

2.2.f [Y/N/M]

Let  $G$  be connected. Then, any Fiedler vector introduces a graph cut. The cut is unique if the Fiedler value is simple.

Shi and Malik introduced in 1999-2000 the use of the normalized cut for image segmentation.

2.2.g [Y/N/M]

Let  $G$  be connected. Let  $\hat{L} = Q\Lambda Q^\top$  be the eigenvalue decomposition. Let  $\Lambda_{\text{sub}}$  be a subset of non-zero eigenvalues. Then,  $Q(:, \Lambda_{\text{sub}})\sqrt{\Lambda_{\text{sub}}}$  can be used for vertex embedding.

2.2.h Optional.

For a connected graph  $G$ , if  $\lambda_{\max}(\hat{L}) < 2$ , then  $\sum_{k=0}^{\infty} c_k P_c^k$ , with  $|c_k| \leq 1$ , converges to a nonzero matrix,

## Part II

# Experiment

This part is for planning and preparing a team project, due a week before the preliminary project report.

E-1 Get a (meaningful) dataset, with datum attributes relevant to intended properties to analyze. Convert the attributes to feature vectors of certain dimension (treated as a hyperparameter). Provide a brief description of the data and the feature extraction.

E-2 Construct an all-to-all pairwise feature interaction matrix (weighted). Provide a brief description of the particular quantity used for the pairwise interaction matrix construction.

E-3 Let  $k$  be a small positive value in a prescribed range such as  $[2^1, 2^6]$ . Find the kNN matrix/graph from the pairwise feature interaction matrix. Find the number of connected components in each kNN graph. Plot the number of connected components across the range of  $k$  values.

Provide a brief description of the process from the all-to-all pairwise interaction matrix to the adjacency matrix of a kNN graph.

Note: If your feature vectors are in a low dimensional geometric space, you may use rNN graphs as well.

E-4 Combinatorial connectivity analysis. For each large connected component  $G$ , plot its degree distribution, and the local cluster coefficient distribution. Identify or describe each distribution type.

E-5 Laplacian spectral analysis. For each large connected component  $G$ ,

- Form its normalized Laplacian  $\hat{L}$ ,  
If the kNN graph is connected after certain value of  $k_c$ , observe the variation in the Fiedler value,  $\lambda_2(k)$  as  $k$  increases,  $k \geq k_c$ .
- Embed its vertices in a three-dimensional invariant subspace of  $\hat{L}$ ,
- Let  $\ell$  be a modest positive integer, e.g.,  $\ell \approx 2^5$ . Find and plot the  $\ell$  smallest eigenvalues of  $\hat{L}$ . Locate a nonzero simple eigenvalue  $\lambda_k$  that is well separated from the rest.  
Partition the graph by the nodal domains of the eigenvector  $q_k$  associated with  $\lambda_k$ . Label/color each vertex by the nodal-domain cluster it belongs to. Color the vertices in a 3D embedding space spanned by  $q_k$  and two other eigenvectors.
- Observe any relationship between the nodal-domain clusters and the original data.

E-6 Optional. Use other graph clustering/segmentation methods.

E-7 Provide a brief discussion on observed key properties and/or initial/preliminary finding.